

# THE ZERO PARADOX

## ZP-J Wheel Addendum

*The Wheel of Fractions is a Wheel*  
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A wheel (Carlström 2001) is an algebraic structure that extends a commutative ring by making division a total operation: every element, including 0, has a reciprocal  $/x$ , so  $/0$  becomes a defined first-class element rather than an error. The two elements this produces —  $\infty = /0$  (the reciprocal of zero) and  $\perp = 0./0$  (an absorbing "undefined" element) — are what distinguish a wheel from a field. ZP-J Self-Reference left open which structure the Zero Paradox porthole gives rise to: a wheel, in which  $\infty$  and  $\perp$  are distinct, or a meadow, in which they collapse. This addendum settles that question.

The main result is `WheelFrac.instWheel (ZPJ_WheelFrac.lean)`: for any commutative ring  $A$  and any multiplicative submonoid  $S$ , the wheel of fractions  $\odot_S A = (A \times A)/\equiv_S$  satisfies every axiom of Carlström's Definition 1.1. The companion result `WheelFrac.inf_ne_bot` shows that, whenever  $0 \notin S$ , the two special elements stay distinct ( $\infty \neq \perp$ ) — so the construction is a wheel, not a meadow. The construction is Carlström's; the contribution here is a faithful, machine-verified encoding of it that is also free of the axiom of choice (footprint `[propext, Quot.sound]`), situated as the algebraic form of the ZP porthole.

### Section I: The Wheel Axioms (Carlström Definition 1.1)

A wheel is a set  $W$  with two binary operations  $+$  and  $\cdot$ , a unary involution  $/$  (reciprocal), and two constants  $0$  and  $1$ , subject to eight axioms. Carlström's Definition 1.1 packages the additive and multiplicative parts as commutative monoids; the ZP Wheel typeclass unbundles those two monoid axioms into their separate equational laws, giving 14 fields that are equivalent to Carlström's eight. The unbundling is bookkeeping only — no axiom is added, removed, or weakened.

#### Typeclass: Wheel (ZPJ\_Wheel.lean)

```
class Wheel (W : Type*) where
  wadd, wmul : W → W → W -- + and ·
  winv : W → W -- the involution /
  wzero, wone : W -- 0 and 1
  -- W1–W3: (W, +, 0) commutative monoid [Carlström (1)]
  -- W4–W6: (W, ·, 1) commutative monoid [Carlström (2), monoid part]
```

### Typeclass: Wheel (ZPJ\_Wheel.lean)

```
-- W7:  $/(/x) = x$  [Carlström (2), involution]
-- W8:  $/(x \cdot y) = /x \cdot /y$  [Carlström (2), involution]
-- W9: weakened distributivity [Carlström (3)]
-- W10:  $(x \cdot /y + z) + 0 \cdot y = (x + y \cdot z) \cdot /y$  [Carlström (4)]
-- W11:  $0 \cdot 0 = 0$  [Carlström (5)]
-- W12:  $(x + 0 \cdot y) \cdot z = x \cdot z + 0 \cdot y$  [Carlström (6)]
-- W13:  $/(x + 0 \cdot y) = /x + 0 \cdot y$  [Carlström (7)]
-- W14:  $x + 0 \cdot /0 = 0 \cdot /0$  [Carlström (8)]
```

The last group of axioms is what makes division total. W7 and W8 make  $/$  an involution that distributes over multiplication; W9–W14 govern how the two derived elements  $\infty = /0$  and  $\perp = 0 \cdot /0$  interact with  $+$  and  $\cdot$ . A wheel in which  $\infty = \perp$  is exactly a meadow: the distinction between the two is the whole content of "wheel, not meadow."

## Section II: The Wheel of Fractions $\odot_S A$

The wheel of fractions generalises the field-of-fractions construction so that it survives division by zero. Start from a commutative ring  $A$  and a multiplicative submonoid  $S \subseteq A$  (a set containing 1 and closed under multiplication). Form pairs  $(x, y) \in A \times A$  — read as the formal fraction  $x/y$  — and quotient by the relation  $\equiv_S$  below.

### Construction (Carlström 2001, pp. 4-5)

$$\odot_S A = (A \times A) / \equiv_S$$

$$(x, y) \equiv_S (x', y') \Leftrightarrow \exists s, s' \in S, s \cdot x = s' \cdot x' \wedge s \cdot y = s' \cdot y'$$

$$0 = [0, 1] \quad 1 = [1, 1]$$

$$[x, y] + [x', y'] = [x \cdot y' + x' \cdot y, y \cdot y']$$

$$[x, y] \cdot [x', y'] = [x \cdot x', y \cdot y']$$

$$/[x, y] = [y, x] \text{ (the involution is pair-swap)}$$

$$\text{Then } /0 = [1, 0] = \infty \text{ and } 0 \cdot /0 = [0, 0] = \perp.$$

The choice of relation matters. The naive cross-multiplication  $x \cdot y' = x' \cdot y$  — the one that defines equality of ordinary fractions — is not an equivalence relation on a general commutative ring: transitivity fails without

a cancellation law. The submonoid-quotient relation  $\equiv_S$  repairs this by witnessing each identification with elements of  $S$ , and it is provably reflexive, symmetric, and transitive (`WheelFrac.srel`). Each of the five operations is then well-defined on the quotient — the proofs that they respect  $\equiv_S$  are the bulk of the formalisation.

## Section III: $\odot_S A$ is a Wheel

With the operations shown well-defined, every one of the 14 typeclass fields is discharged. The additive and multiplicative monoid laws and the distributivity and porthole axioms all reduce, after lifting representatives, to ring identities in  $A$  closed by ring normalisation; the two involution laws (W7, W8) hold definitionally because pair-swap is its own inverse and commutes with the componentwise product.

### Theorem: `WheelFrac.instWheel (ZPJ_WheelFrac.lean)`

$\forall \{A : \text{Type}^*\} [\text{CommRing } A] (S : \text{Submonoid } A),$

$\text{Wheel } (\odot_S A)$

For every commutative ring  $A$  and multiplicative submonoid  $S$ , the wheel of fractions  $\odot_S A$  satisfies all 14 fields of the ZP Wheel typeclass — equivalently, all eight axioms of Carlström’s Definition 1.1.

Sorry-free. Lean purity: [propext, Quot.sound] — Classical.choice-free.

## Section IV: Wheel, Not Meadow

A wheel collapses to a meadow precisely when its two special elements coincide. For the wheel of fractions this collapse happens exactly when  $0 \in S$ : if  $S$  contains a zero divisor witness for 0, the fraction  $[1,0]$  and the fraction  $[0,0]$  become identified. The natural hypothesis  $0 \notin S$  (which holds whenever  $S$  is the complement of a prime ideal, the usual case) keeps them apart.

### Theorem: `WheelFrac.inf_ne_bot (ZPJ_WheelFrac.lean)`

$\forall \{A : \text{Type}^*\} [\text{CommRing } A] (S : \text{Submonoid } A),$

$(0 : A) \notin S \rightarrow \infty \neq \perp \text{ in } \odot_S A$

When  $0 \notin S$ , the reciprocal of zero ( $\infty = /0$ ) and the absorbing element ( $\perp = 0 \cdot /0$ ) are distinct. The construction is a genuine wheel, not a meadow.

Proof:  $\infty = \perp$  would yield witnesses  $s, s' \in S$  with  $s \cdot 1 = s' \cdot 0$ , forcing  $s = 0$  and hence  $0 \in S$  — contradicting the hypothesis.

Sorry-free. Lean purity: [propext, Quot.sound] — Classical.choice-free.

### Remark — The Porthole Connection

In ZP-J, the porthole is the point where  $\text{val}(\perp) = \infty$  and  $\perp = \{\perp\}$  coincide — the same structural fact written in the 2-adic valuation ( $v_2(0) = \infty$ ) and in ZF+AFA (the Quine atom). The wheel of fractions is the algebraic face of that point:  $/0$  is defined and distinct from the absorbing  $\perp$ , exactly the behaviour the porthole predicts. The concrete carrier `ZPWheelElem` (`ZPJ_Wheel.lean` §III–VI) makes this explicit on the rationals extended with  $\infty$  and  $\perp$ , where  $\text{val}(x) = \infty \Leftrightarrow /x = \infty$  is proved directly (`zpw_top_val_iff_inv_is_inf`). This addendum’s headline result is the general construction over an arbitrary commutative ring; the concrete carrier is the illustrative special case.

## Section V: Scope, Purity, and Relationship to Carlström

Two scope boundaries are worth stating plainly. This is a formalisation, not new mathematics. The wheel of fractions and the theorem that it is a wheel are Carlström’s. What is contributed here is a machine-checked encoding faithful to Definition 1.1, with a verified axiom footprint, placed in the ZP porthole context. Ring structure is an input, not a conclusion. The construction starts from a commutative ring and a submonoid; it does not derive wheel structure from the ZP lattice axioms alone. The bridge typeclass that would state such a derivation, `WheelValuationStructure` (`ZPJ_Wheel.lean` §VII), is defined but its porthole condition  $\text{val}(0) = \top$  is an assumed axiom, motivated by the ZP argument rather than type-checked as necessary.

### Lean Source Files

`ZPJ_WheelFrac.lean` — `rel`, `srel`, the five quotient operations (`waddF`, `wmulF`, `winvF`), `instWheel`, `inf_ne_bot`. The headline results of this addendum.

`ZPJ_Wheel.lean` — the `Wheel` typeclass (Carlström Def 1.1, 14 fields), the derived elements `wheelInf` / `wheelBot`, the concrete carrier `ZPWheelElem`, and the porthole correspondence `zpw_top_val_iff_inv_is_inf`.

Both files in `ZeroParadox/` in the public repository.

### Axiom Footprint (headline results: `instWheel`, `inf_ne_bot`)

[`propext`, `Quot.sound`] — `Classical.choice-free`.

`propext` — propositional extensionality (standard in Lean 4)

`Quot.sound` — quotient soundness (standard in Lean 4; the construction is a quotient, so this is expected and unavoidable)

No `Classical.choice`. No `Dependent Choice`. No set-theoretic assumptions.

### Remark R-J.W — Relationship to Carlström's Theorem

Carlström introduced wheels in *Wheels — On Division by Zero* (Research Reports in Mathematics No. 11, Department of Mathematics, Stockholm University, 2001; a Licentiate thesis), where Definition 1.1 (p. 5) gives the eight wheel axioms and the wheel of fractions is constructed (§1.2, §4.2). The work was later published as *Wheels — on division by zero*, *Mathematical Structures in Computer Science* 14(1):143–184, 2004. Carlström proved there that the wheel of fractions of a commutative ring is a wheel. The result here is that theorem, encoded in Lean 4 against a typeclass that reproduces his Definition 1.1 field for field, and discharged without the axiom of choice. The encoding is what is new: a third party can read the 14 fields against Carlström's eight axioms and confirm the correspondence, and can read the axiom footprint to confirm the choice-free claim. Whether the porthole condition  $\text{val}(0) = \top$  can be derived — rather than assumed — from upstream ZP structure remains the open question flagged in `ZPJ_Wheel.lean` §VII–VIII.

Endnote: This document is an addendum to ZP-J Self-Reference and reads after it. ZP-J established the porthole ( $\text{val}(\perp) = \infty$ ,  $\perp = \{\perp\}$ ); this document gives its algebraic form, the wheel of fractions, and the machine-verified proof that the construction is a wheel rather than a meadow. All results sorry-free in Lean 4 as of June 2026, footprint [propept, Quot.sound].